

# Kaden Musard

**Embedded Systems Engineer | Full-Stack Developer | Robotics Specialist**

Website: <https://www.kadens.tech/> | Email: [kadenoncomputation@gmail.com](mailto:kadenoncomputation@gmail.com) |

Phone: 240-385-5252 | GitHub: <https://github.com/kaden-code>

## Professional Summary

Results driven Embedded Systems Engineer and Full Stack Developer with 4+ years of hands on experience in robotics, IoT systems, computer vision, full stack development, and real time control systems. Proven track record of designing and implementing complex systems from concept to deployment, including autonomous vehicles, IoT enabled hardware modifications, ESP32 flight control systems, WebSocket dashboards, and responsive portfolio applications. Expertise in C++, Python, TypeScript, JavaScript, React, PCB design, sensor fusion, PID control, and real time embedded systems. Skilled in bridging hardware software domains to create innovative, real world solutions.

## Technical Skills

### Programming Languages

C++, Python, TypeScript, JavaScript (ES6+), HTML5, CSS3, Arduino C++, Embedded C++

### Frameworks & Libraries

React, Vite, Tailwind CSS, Framer Motion, React Router, OpenCV, MediaPipe, ESPAsyncWebServer, WebSockets, ArduinoJson, NumPy, Pandas, face\_recognition, LittleFS

### Embedded Systems & Hardware

Arduino, ESP32, IMUs, ultrasonic sensors, IR receivers, DHT11 temperature/humidity sensors, servo motors, stepper motors, DC motors, brushless motors, ESCs, RC receivers, PWM control, I2C/SPI/UART communication protocols, GPIO programming, LittleFS storage

## **Electronics & Circuit Design**

Understanding of electrical fundamentals, PCB design and layout, precision soldering, circuit prototyping, MOSFET drivers, power distribution systems, sensor integration, schematic capture, breadboard prototyping, voltage regulation, current-limiting circuits

## **Computer Vision & AI**

Real time object detection, face detection and recognition, hand tracking, Haar cascades, MediaPipe pose estimation, feature extraction, image processing, matrix manipulation, background subtraction, motion detection, camera calibration

## **Control Systems & Robotics**

Real time control systems, autonomous navigation algorithms, obstacle avoidance, motor control calibration, interrupt driven signal processing, sensor fusion, complementary filtering, PID control theory, cascaded PID stabilization

## **Web Development & IoT**

Full stack development (HTML/CSS/JavaScript/TypeScript), React component architecture, responsive design, Tailwind CSS styling, WebSocket real time communication, AsyncWebServer implementation, React Router navigation, Vite production builds, Vercel deployment workflow, SoftAP configuration, JSON protocol design, DOM manipulation, event driven programming, PROGMEM optimization, cross platform compatibility (mobile, desktop, etc)

## **Development Tools & Practices**

Git/GitHub version control, serial debugging, multithreaded programming, memory optimization, real time performance tuning, modular code architecture, state machine design, interrupt handling, non-blocking event loops, mobile-first debugging, production deployment troubleshooting

# Professional Experience

## Independent Embedded Systems Developer | Self-Directed Projects

*November 2022 - Present*

### Interactive Engineering Portfolio Website (Aug 2026)

Website: <https://www.kadens.tech/>

Check this project out on GitHub: <https://github.com/kaden-code/PORTFOLIO-SITE>

- Built a fully responsive React and TypeScript portfolio website showcasing embedded systems, robotics, computer vision, IoT, and full-stack projects
- Developed a polished project carousel with Previous, Next, Shuffle, direct jump controls, category filters, and animated transitions
- Replaced native project selection with a custom mobile friendly dropdown and fixed mobile overflow issues across responsive layouts
- Created a dedicated contact page with resume download, GitHub, LinkedIn, email, and phone links
- Configured production ready Vite/Vercel deployment and solved React Router single-page application routing issues
- **Technologies:** TypeScript, JavaScript, HTML, CSS, React, Vite, Tailwind CSS, Framer Motion, Lucide React, React Icons, React Router (LOC pending)

## Drone Flight Controller + PID Tuning Website (July 2025)

Check this project out on GitHub: <https://github.com/kaden-code/pidFlightControllerSite>

- Built a full stack ESP32 based quadcopter flight controller in Arduino C++ with real time sensor fusion, dual cascade PID stabilization, and live telemetry
- Designed interrupt driven RC signal reading for six channels and maintained a strict 4 ms control cycle for flight stabilization
- Calibrated MPU6050 gyro/accelerometer bias and tuned a complementary filter for stable attitude estimates
- Implemented persistent PID gain tuning with LittleFS and ArduinoJson while avoiding blocking operations during flight control
- Developed a responsive WebSocket driven dashboard for live telemetry and configuration from a browser
- Fabricated and integrated hardware including ESP32, MPU6050, brushless motors, ESCs, RC receiver, custom PCB/perf board, and 3D printed chassis
- **Technologies:** C++, HTML/CSS/JavaScript, ESP32, Arduino, Wire, ESP32 Servo, ESPAsyncWebServer, WebSocketsServer, LittleFS, ArduinoJson, PID control, sensor fusion, PCB design, soldering, 3D-printed chassis design

## IoT LED Control System for Electric Scooter (Sept-Oct 2024)

Check this project out on GitHub: <https://github.com/kaden-code/esp32ScooterLedMod>

Demo video: <https://www.youtube.com/watch?v=CxjpcTXjc0g>

- Built standalone Wi-Fi hotspot system allowing real time customization of dual RGB LED channels via a browser-based interface
- Embedded 592 line HTML/CSS/JavaScript interface into PROGMEM, achieving zero runtime RAM overhead
- Implemented custom WebSocket protocol with ArduinoJson parsing for color sequences and independent left/right channel control
- Designed PCB with 6 MOSFET drivers for voltage step up and soldered complete circuit integration into scooter frame
- Developed non-blocking timing loop for dynamic color sequences with concurrent WebSocket event handling
- **Technologies:** C++, HTML/CSS/JavaScript, ESP32, AsyncWebServer, WebSockets, ArduinoJson, PWM, PROGMEM, PCB design, MOSFET drivers

## Computer Vision Applications Suite (Jan-June 2024)

Check this project out on GitHub (Face recognition): <https://github.com/kaden-code/fileSystemFaceRecognition>

Demo video (Face recognition): <https://www.youtube.com/watch?v=mxXeD08ib00>

- Developed real time facial recognition system with live enrollment, achieving 128 dimensional face embeddings and distance based matching with optimized threshold tuning
- Created dual mode hand tracking Pong games (1P and 2P) using MediaPipe, implementing handedness classification and collision detection with real-time physics
- Built comprehensive OpenCV project portfolio including object detection, hand tracking, and image processing with 2000+ lines of Python code
- Optimized real time performance through frame resizing, coordinate scaling, and memory efficient encoding management
- **Technologies:** Python, OpenCV, MediaPipe, NumPy, Pandas, face\_recognition, Haar cascades, real-time video processing

## Autonomous Obstacle Avoidance Robot Car (Dec 2023)

Check this project out on GitHub: <https://github.com/kaden-code/arduinoRobotCar>

Demo video: <https://www.youtube.com/watch?v=S3dMc1q6CSc>

- Engineered dual-mode robotic car with IR remote control and autonomous navigation using servo mounted ultrasonic sensor
- Implemented real time decision making algorithm analyzing multiple sensor scans to determine optimal path and execute precise 90° turns
- Calibrated motor control for consistent turning radius and optimized scan algorithms to balance decision latency with accuracy
- Developed fail safe routines and signal interference mitigation between IR receiver and ultrasonic sensor
- **Technologies:** C++, Arduino Uno R3, HC-SR04 ultrasonic sensor, TB6612 motor driver, IR receiver, servo motor, PWM control

## 3D👁️MIND Visualization Web Application (April 2023)

Check this project out on GitHub: <https://github.com/kaden-code/visualizationApp>

Demo video: <https://www.youtube.com/watch?v=n47hyg3BLVo>

- Built full featured single page web application with three interactive visualization exercises (Image Roulette, Image Description, Image Streaming)
- Implemented complex timer orchestration with nested setTimeout/setInterval calls and synchronized audio cues
- Developed dynamic DOM lifecycle management with custom list creation, validation, and real time integration into game modes
- Created fully responsive design from desktop to mobile using CSS Flexbox and extensive media queries
- **Technologies:** HTML5, CSS3, Vanilla JavaScript, responsive design, DOM manipulation, event-driven programming (1,401 total LOC)

## Hand-Tracking Pong Game (May - June 2024)

Check this project out on GitHub: <https://github.com/kaden-code/computerVisionPong>

Demo video: <https://www.youtube.com/watch?v=oXxv7skM74w>

- Developed interactive Pong game controlled entirely by hand gestures using computer vision and MediaPipe
- Implemented real time hand tracking with collision detection and physics based ball movement
- Created dual mode gameplay supporting both single-player and two-player configurations with handedness classification
- Optimized frame processing for smooth gameplay with minimal latency
- **Technologies:** Python, OpenCV, MediaPipe, real-time video processing

## Arduino Robotics Foundation Projects (Oct - Dec 2023)

- Completed comprehensive electronics and robotics curriculum with 5,000+ lines of C++ across multiple sensor and actuator projects
- Mastered I2C/SPI/UART communication protocols, serial debugging, and real-time embedded programming paradigms
- Integrated diverse components including LCD displays, temperature sensors, IR modules, stepper motors, and relay systems
- Developed modular, reusable code architecture for sensor integration and hardware abstraction
- **Technologies:** C++, Arduino, Wire Library, LiquidCrystal, Servo, DHT, IRremote, multiple sensor/actuator modules



# Education & Self-Directed Learning

**Self-Taught Developer** | November 2022 - Present

- Systematically progressed from Python fundamentals through web development (HTML/CSS/JavaScript) to embedded systems and robotics
- Completed comprehensive learning path: Python → HTML/CSS → JavaScript → Arduino C++ + electrical theory → ESP32 → Computer Vision
- Developed 15+ substantial projects demonstrating incremental skill building and cross disciplinary integration
- Applied meta learning techniques using custom visualization tools to enhance programming skill acquisition

## Key Projects Summary

- **Interactive Engineering Portfolio Website** - Responsive React/TypeScript portfolio with carousel, filters, animations, contact page, resume download, and Vercel-ready routing (LOC pending)
- **Drone Flight Controller + PID Tuning Website** - ESP32 quadcopter controller with sensor fusion, cascaded PID stabilization, WebSocket telemetry, persistent tuning, and custom hardware integration (C++ 1,029 LOC; HTML/CSS/JS 852 LOC)
- **Scooter LED Modification** - IoT enabled RGB lighting system with WebSocket control and custom PCB (841 LOC)
- **Facial Recognition System** - Real time identification with live enrollment and 128D embeddings (115 LOC)
- **Hand-Tracking Pong Game** - Computer vision controlled game with dual-mode gameplay (274 LOC)
- **Autonomous Robot Car** - Dual mode obstacle avoidance with servo-mounted ultrasonic scanning (614 LOC)
- **3D👁️MIND Web App** - Full featured visualization training platform with responsive design (1,401 LOC)

- **Computer Vision Suite** - Hand tracking games and object detection applications (2,000+ LOC)
- **Arduino Robotics Portfolio** - Comprehensive sensor/actuator integration projects (5,000+ LOC)

## Additional Skills

- **Problem Solving:** Systematic debugging of complex hardware software interactions, optimization under resource constraints, real time performance tuning
- **System Design:** End to end architecture planning, modular code organization, scalable embedded systems design, full-stack IoT architecture
- **Hardware Fabrication:** PCB layout and fabrication, precision soldering, 3D designs for mechanical integration, custom chassis design, component selection and sourcing
- **Documentation:** Comprehensive project documentation, code commenting, technical writing, Detailed video documentation
- **Continuous Learning:** Self directed skill acquisition, rapid technology adoption, cross disciplinary integration, production deployment workflows

## Portfolio & Code Samples

Website: <https://www.kadens.tech/> | GitHub: <https://github.com/kaden-code> - showcasing all substantial projects with detailed README files and in depth video demos

*Available for full-time positions in embedded systems engineering, software engineering, robotics development, IoT systems, or related fields. Open to relocation.*